PLC Ladder Logic Control Systems

Project Overview

This project focused on developing and testing PLC ladder logic programs using the LogixPro PLC Simulator. The work included three automation applications: a door control system, a conveyor and silo control system, and a two-way traffic light control system. Each program was designed to meet specific control requirements using ladder logic instructions such as relays, timers, counters, sensors, switches, output coils, and interlock conditions.

The goal of the projects was to apply logic control concepts to simulated industrial systems and verify that each system responded correctly to user inputs, sensors, and operating conditions. The programs were tested directly in LogixPro using the built-in simulators to confirm correct operation before submission.

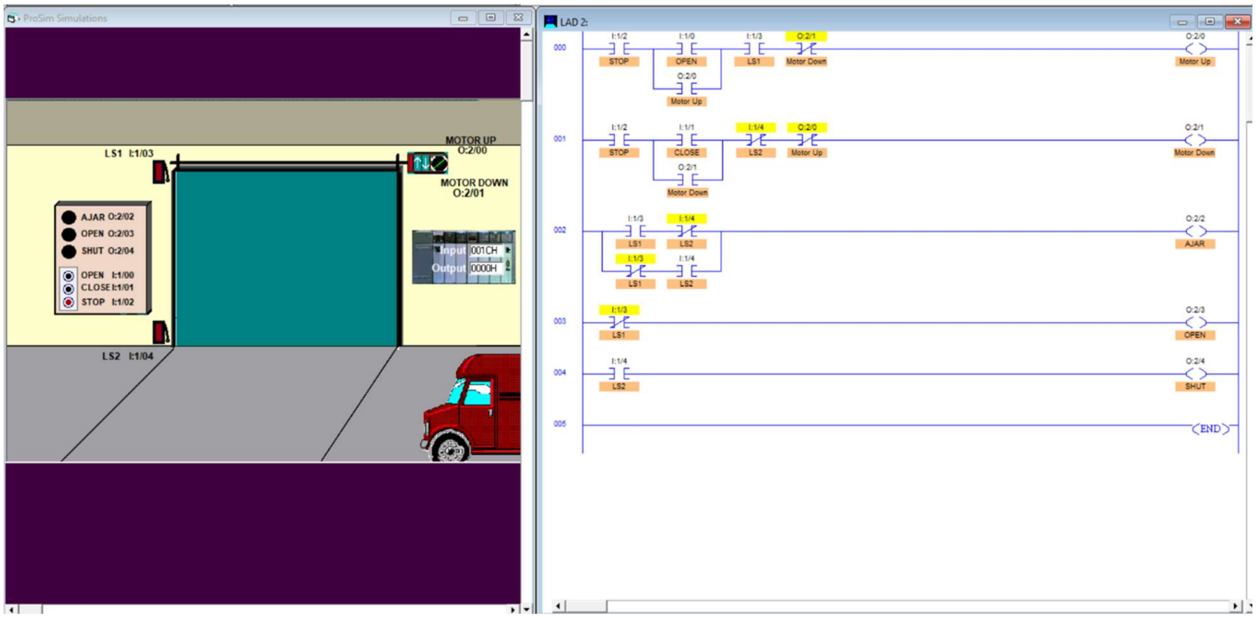
Software and Tools Used

Software: LogixPro PLC Simulator
Concepts Applied: PLC ladder logic, relays, timers, counters, input/output logic, motor control, sensor-based control, interlocks, and status indicator lights
Simulators Used: Door Simulator, Silo Simulator, and Traffic Light Simulator

Project 1: Door Control System

The first PLC program controlled the door simulator in LogixPro. The system was designed to open and close a motorized door while responding correctly to the open, close, and stop switches. The logic also controlled indicator lamps for the open, shut, and ajar door conditions.

The ladder logic was programmed so that pressing the open switch energized the motor up output if the door was not already fully open. Pressing the close switch energized the motor down output if the door was not already fully closed. The stop switch stopped door movement immediately, and the door remained stopped after the switch was released. Limit switches were used to prevent the motor from continuing once the door reached the fully open or fully closed position. Indicator outputs were also programmed so the open lamp turned on when the door was fully open, the shut lamp turned on when the door was fully closed, and the ajar lamp turned on when the door was between both limit positions.



Main requirements completed:

- Programmed open, close, and stop switch logic for a motorized door.
- Used limit switches to prevent motor operation when the door was already fully open or fully closed.
- Added open, shut, and ajar indicator lamp logic based on door position.
- Tested the system in the LogixPro Door Simulator to verify correct motor and lamp behavior.

Project 2: Two-Way Traffic Light Control System

The third PLC program controlled a two-way traffic light simulator in LogixPro. The system was designed to create a timed traffic sequence where each direction alternated between red, green, and yellow lights while keeping one light active in each direction at all times.

The ladder logic used timer instructions to control the light sequence. In one direction, the red light stayed on while the opposite direction cycled through green and yellow. After the timing sequence was complete, the opposite direction repeated the same pattern. The traffic cycle used a 7-second red phase, a 5-second green phase, and a 2-second yellow phase. The program was tested to verify that both directions followed the required timing sequence and that no unsafe light condition occurred.

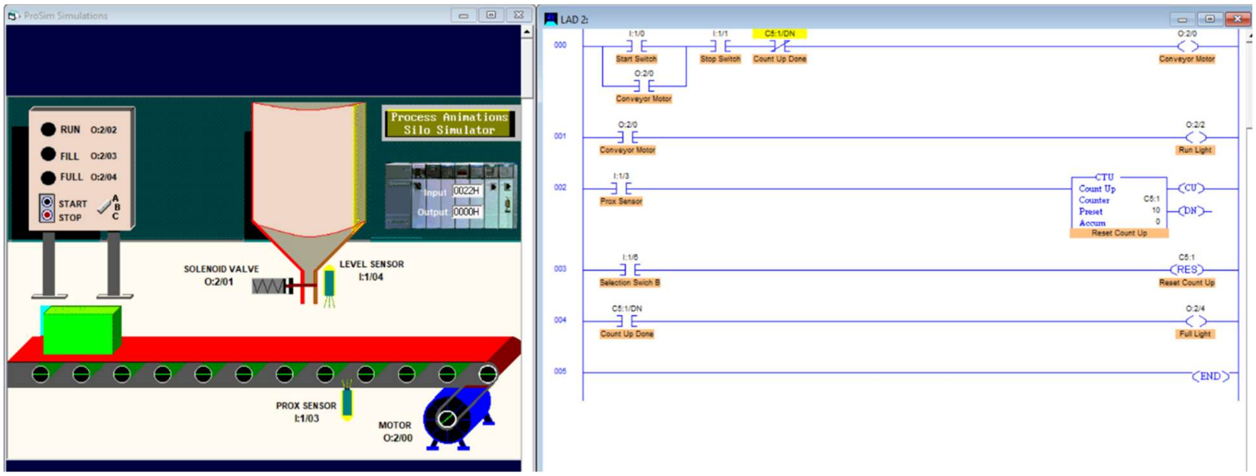
Main requirements completed:

- Programmed timed red, green, and yellow light sequences for two-way traffic control.
- Used timer instructions to create 7-second, 5-second, and 2-second operating intervals.
- Verified that one light remained active in both traffic directions at all times.
- Tested the full sequence in the LogixPro Traffic Light Simulator.

Project 3: Conveyor and Silo Control System

The second PLC program controlled the conveyor and silo simulator in LogixPro. The system was designed to start and stop the conveyor motor, count boxes using a proximity sensor, and stop the conveyor after the box count reached 10.

The ladder logic used a start switch to energize the conveyor motor and a stop switch to turn it off. A run light was programmed to turn on whenever the conveyor motor was active. A proximity sensor was used with a counter instruction to count each box passing through the system. Once the count reached 10, the conveyor motor stopped automatically, and the full light turned on. A selector switch was used to reset the box count back to zero, allowing the system to restart the counting process.



Main requirements completed:

- Programmed start and stop switch logic for conveyor motor control.
- Used a proximity sensor and counter instruction to count boxes.
- Stopped the conveyor automatically when the count reached 10.
- Added run and full indicator lights to show system operating status.
- Programmed reset logic using the selector switch to clear the box count.

Conclusion

The PLC ladder logic projects provided hands-on practice with automation and control systems in LogixPro. By programming door control, conveyor/silo control, and two-way traffic control simulations, I gained experience with motor control, sensor-based counting, timed sequences, reset logic, interlocks, and indicator outputs. These projects helped connect mechanical controls concepts to practical PLC applications commonly used in manufacturing and industrial automation.